DATAVIZON
TECHNOLOGY - TAX - FINANCE
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Clear My Interview

AI-POWERED

FULL STACK JAVA

Trainer- KASI YEDUMATI



FULL STACK JAVA 2026-27

KSR DATAVIZON
TECHNOLOGY -TAX - FINANCE
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About KSR Datavizon

Welcome to KSR Datavizon, we are your one-stop solution for all trending technologies. We provide in-depth knowledge about all trending courses with 100% placement assistance.

You also will get hands-on demonstrations, interactive live sessions to learn and grow more on current trending courses like AI, Machine Learning, Data Science, Cloud Computing, Full stack Development, Cyber Security, Credit Risk modelling, DevOps, Azure, AWS, Power BI, Web Development, Data Engineer, ADF, MSBI and Snowflake and many more. You also get career guidance, soft skills sessions and resume building and all other interview tips to get settled in your dream career.

In this course, you'll master the complete journey to becoming a Full Stack Java Developer — starting from core Java fundamentals, object-oriented programming, and data structures to frontend development with HTML, CSS, and JavaScript, building dynamic UIs with React, and backend development using Spring Boot.

What you will get !



Course Completion Certificate



Live Interactive session



End-to-End Full Stack Projects



Free Career Guidance



Hands-on Labs with Latest Java, Spring & AI Tools



LMS Platform Access for 365 Days for Recorded Sessions

Programming Foundations

Month 1

Goal: remove fear of coding.
No prior coding experience needed. We start from zero.

So, start slowly but strongly.

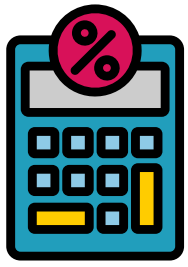
Topic

- Programming mindset
- Java 17 setup (JDK, IDE)
- Variables
- Data types
- Operators
- If/else
- Loops

- **Practice problems**
  - even/odd
  - prime number
  - factorial
  - palindrome
  - multiplication table

Assignment

- **Build programs:**
  - simple calculator
  - number guessing game



Programming Foundations

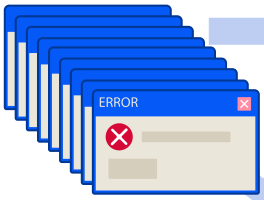
Month 1

Topic

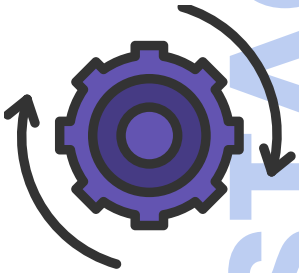
- Arrays
- Functions
- String basics



- **Debugging basics:**
  - compile errors
  - runtime errors
  - logic errors



- **Practice problems**
  - reverse array
  - find max element
  - count vowels
  - string reverse



Assignment

- **Student Marks Analyzer**



FULL STACK JAVA

OOP, Collections & Git

Month 2

Topic

- OOP fundamentals
- Classes & Objects
- Abstraction & Interfaces
- Constructors
- Encapsulation
- Inheritance
- Polymorphism

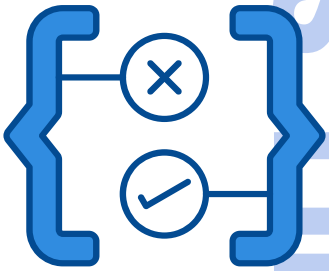


- **Practice problems**
  - bank account class
  - employee class
  - vehicle hierarchy



Topic

- Collections
- ArrayList
- HashMap
- HashSet
- Exception handling
- Custom exceptions
- File handling (CSV / text)





FULL STACK JAVA

OOP, Collections & Git

Month 2

Mini Project

Student Management CLI



- **Features**
  - add student
  - delete student
  - update student
  - search student
  - display students

Students will push this to GitHub.

- **Developer tools: Git & GitHub**
  - git init, add, commit
  - Push & clone on GitHub
  - Branching basics
  - Daily commit workflow



FULL STACK JAVA

Database

Month 3

SQL

1. Setup & Connection



- Install MySQL (Windows / macOS / Linux)
- MySQL Workbench
- Connect via CLI & Workbench
- Create training_db, useful commands, troubleshooting

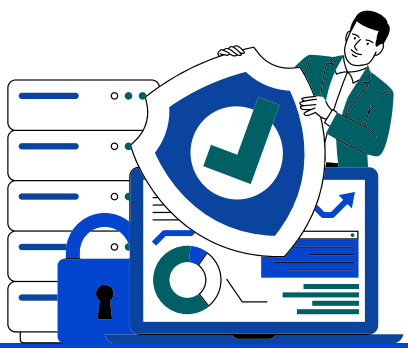
2. Introduction to Databases



- Data, database, DBMS
- Relational vs non-relational
- Tables, rows, columns
- DBMS vs RDBMS

3. SQL Basics (DDL / DML / DQL)

- SQL language categories: DDL, DML, DQL, TCL, DCL
- DDL vs DML commit behavior
- DDL: CREATE, data types, constraints, DESCRIBE, ALTER, DROP, TRUNCATE
- DML: INSERT, UPDATE, DELETE
- DQL: SELECT (columns, *, WHERE)
- DCL commands



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Database

Month 3

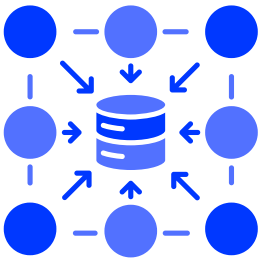
SQL

4. Querying with SQL



- Comparison operators, IS NULL / IS NOT NULL
- LIKE (wildcards)
- AND / OR / NOT
- IN, BETWEEN
- ORDER BY, DISTINCT
- Pagination: LIMIT, OFFSET

5. Aggregations & GROUP BY



- COUNT, SUM, AVG, MIN, MAX
- Alias
- GROUP BY, WHERE + GROUP BY, HAVING

6. Common Concepts

- SQL expressions
- String, date, arithmetic, CAST functions
- Set operations: UNION / UNION ALL (ORDER BY, pagination)

7. Modeling Databases



- ER model: entity, attributes, keys, relationships
- Cardinality (1:1, 1:N, M:N)
- Primary key, foreign key
- Constraints & creating related tables

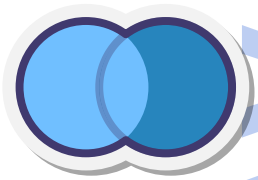
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Database

Month 3

SQL

8. Joins



- Natural, INNER, LEFT, RIGHT, FULL, CROSS, Self join
- Multi-table joins
- Joins with filters & aggregations

9. Views & Subqueries



- CREATE / query / DROP view
- Subqueries in WHERE, SELECT, FROM
- Correlated subquery, EXISTS / NOT EXISTS
- Subquery vs JOIN

10. Transactions, DCL & Indexes

- ACID (Atomicity, Consistency, Isolation, Durability)
- TCL: START TRANSACTION, COMMIT, ROLLBACK, SAVEPOINT
- DCL: GRANT, REVOKE
- Indexes: CREATE / DROP, when to use



11. Window Functions & CTEs

- ROW_NUMBER, RANK, DENSE_RANK
- CTEs (WITH), multiple CTEs, recursive CTEs

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Database

Month 3

SQL

12. Practice Exercises

- Chapter-wise homework aligned to modules above

13. SQL Case-Study Projects

- **AcmeHR — Employee Management**
- **TechMart — E-Commerce**
- **MedCare — Hospital Management**
- **SecureBank — Banking**



JDBC + Web Fundamentals

Month 3

Topic

- Java Database Connectivity
  - JDBC
  - Connection
  - Prepared statements

Project

- Student Management (JDBC version)
- Students connect Java app to MySQL

Topic

- Web fundamentals
  - HTTP
  - REST APIs
  - JSON
- Frontend
  - HTML
  - CSS
  - JavaScript
  - Bootstrap



Java Backend

Month 4

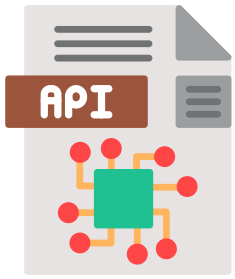
Spring Boot Backend Development.

Build real APIs from day one.

Topic

- Spring fundamentals
- Dependency Injection
- Spring Boot setup (Maven)

- **Controllers**
  - Build API
  - /hello
  - /user
  - /product



- **Spring MVC architecture**
  - Controllers
  - Services
  - Repositories



Project Start

Employee Management System

Java Backend

Month 4

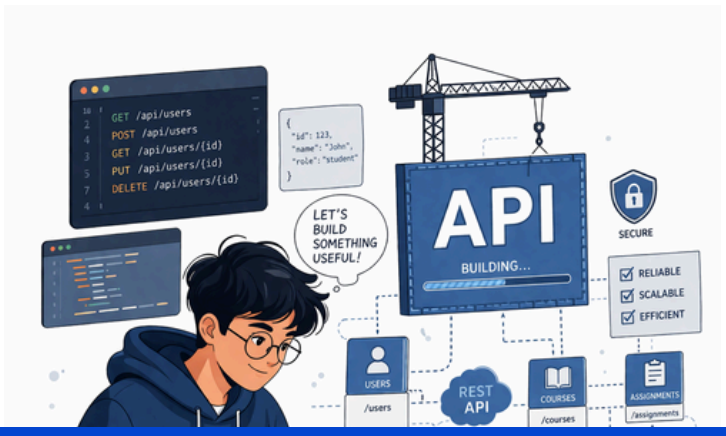
Topic

- Spring Data JPA
- Entities
- Relationships

- **CRUD APIs**
- **Students build APIs**
  - add employee
  - update employee
  - delete employee
  - search employee
- **Topic**
  - Validation & DTOs
  - Global exception handling
  - Spring Security basics
  - JWT login
  - JUnit & Mockito testing



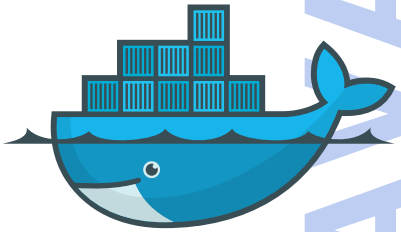
Students test APIs using Postman



Full Stack Integration + Deployment

Month 4

- Docker basics
- Build & Package
  - Maven build lifecycle
  - Executable JAR packaging
  - Spring profiles (dev / prod)
  - Dockerfile for Spring Boot
- Deployment
  - AWS EC2
  - Application hosting





Students deploy:

- Employee Management System
- Live project URL for your portfolio



Cloud Deployment (AWS)

- AWS fundamentals
- EC2 instance setup
- Security groups
- SSH connection
- Deploy Spring Boot application
- java -jar app.jar

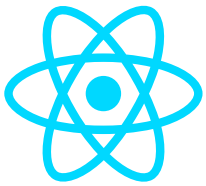


Frontend with React

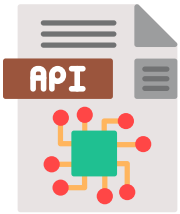
Month 5

REACT

- Introduction to React
- Installation and setup
- Components
- Props
- State management
- Component lifecycle management
- Event handling
- Lists & Keys
- Forms
- Styling
- Routing
- API Integration
- Hooks
- Unit Testing
- Connecting frontend to backend
- Project build and deployment







Mini Project

Library Web App (simple UI)

FULL STACK JAVA

Integration + AI Tools

Month 5

Goal: connect backend with UI and deploy.

Topic

- **AI-Powered Development**
  - AI coding assistants (Copilot, Claude)
  - AI for debugging & unit tests
  - Spring AI: calling LLM APIs

Add an AI feature to your project.

- **React API integration**
  - fetch
  - axios

Students connect React with Spring Boot APIs.

- **Git in a team**
  - feature branches
  - pull requests & code reviews
  - merge conflicts
  - GitHub Issues & project boards

Production Deployment

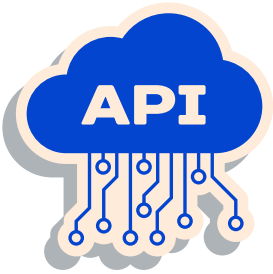
Month 5

- Git branching, PRs & release flow
- Dev / Staging / Production environments
- Environment variables & secrets management
- Docker & Docker Compose
- CI/CD with GitHub Actions
- Nginx reverse proxy & HTTPS
- Cloud deployment (EC2 / managed platforms)
- Logging, health checks & basic monitoring
- Database migrations & backup awareness

- **Configure:**
  - port
  - firewall
  - public access



- **Students deploy:**
  - Capstone project with CI/CD
  - Outcome: live app with HTTPS & CI/CD



- **Example:**

  http://3.110.x.x/api/employees

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Interview + Projects

Month 5

Build, secure and deploy a complete full stack app.

Capstone project

Options

- E-commerce system
- Job portal
- Learning management system



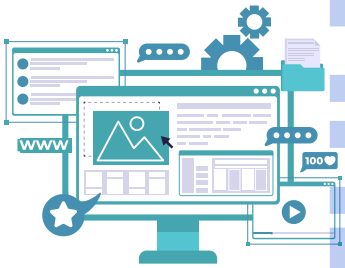
Must include

- Spring Boot backend
- React frontend
- MySQL database
- JWT authentication
- AWS deployment

Final Student Portfolio

Each student must graduate with:

- Student Management CLI
- Library Web App
- Employee Management System
- Capstone Project



All projects on:

- GitHub
- Live URL

DSA FOR JAVA

Weekly

2–3 problems every week, from Month 2 onwards.

Topic

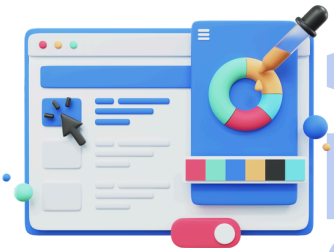
DSA basics



Problems:

- two sum
- reverse string
- valid parentheses
- merge arrays

More DSA



Problems:

- Sliding window
- Hashmap problems
- Duplicate detection

System design basics



Explain:

- how WhatsApp works
- how Instagram works
- how Uber works
- Monolith vs microservices

FULL STACK JAVA

Interview + Projects

Month 5

Top 10 Core DSA Problems

Master these essential problems:

- two sum
- maximum subarray
- valid palindrome
- anagram check
- merge sorted arrays
- move zeros
- stock buy sell
- longest substring
- first unique character
- binary search



Core patterns asked in entry-level interviews

Weekly Debugging Day

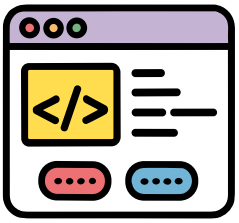
Every Friday:

Fix real broken code, such as:

- NullPointerException
- SQL syntax error
- Spring bean error
- Find the root cause and fix it



Real-world developer skill.



How Each Session Works

Each session:

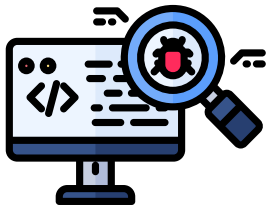
- 5 min lab kit setup
- 25 min concept demo
- 35 min hands-on lab
- 10 min recap + homework



Hands-on practice in every session

What Makes This Program Different

Our core focus:



Debugging skills
Fix real bugs every week.



GitHub portfolio
Every project on GitHub.



Live projects
Recruiters love deployed apps.



Mock interviews
Regular practice.

Program Overview

Program: KSR Job Ready Java Full Stack Developer Program

Our promise:

Become a Java Full Stack Developer with real projects, live cloud deployment, and industry-ready skills.



Live Interactive Classes

Weekly live sessions with real-time Q&A and discussions



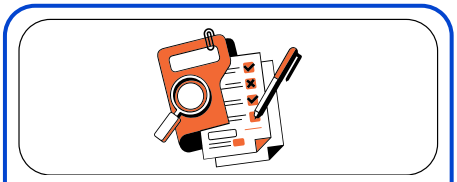
130+ Hours Training

Comprehensive hands-on learning with practical focus



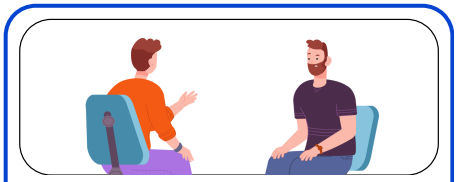
50+ Lab Exercises

Real-world practical labs covering all topics



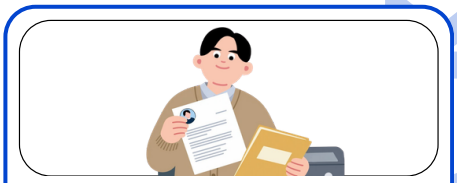
Capstone Projects

Portfolio-ready projects for your resume



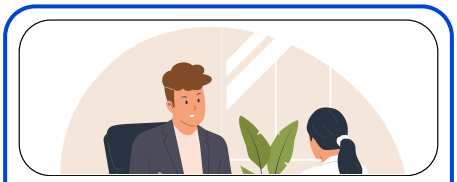
1-on-1 Mentorship

Personal guidance sessions for career development



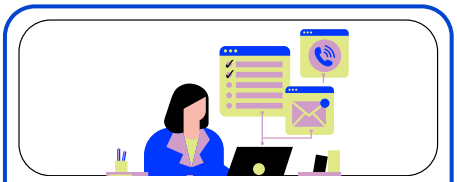
Resume Optimization

LinkedIn profile and resume review and enhancement



Interview Preparation

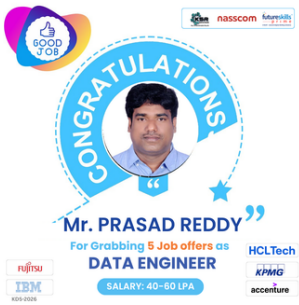
Mock interviews and technical guidance



Job Assistance

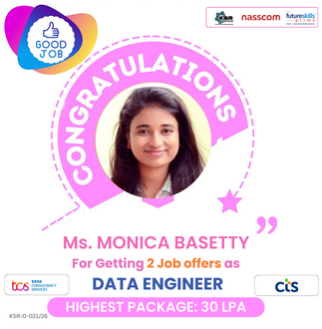
Placement support and job opportunity alerts

ALUMNI PLACEMENTS (ALL KSR PROGRAMS)

































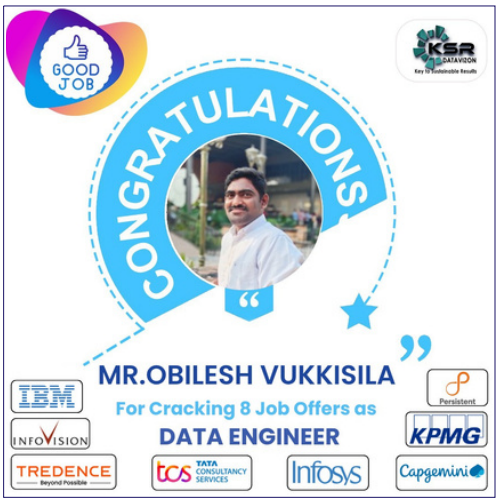


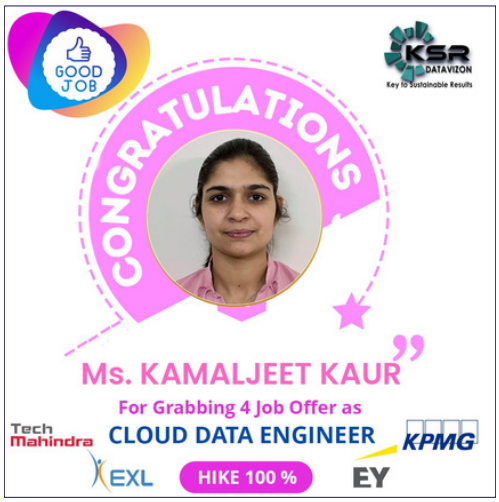


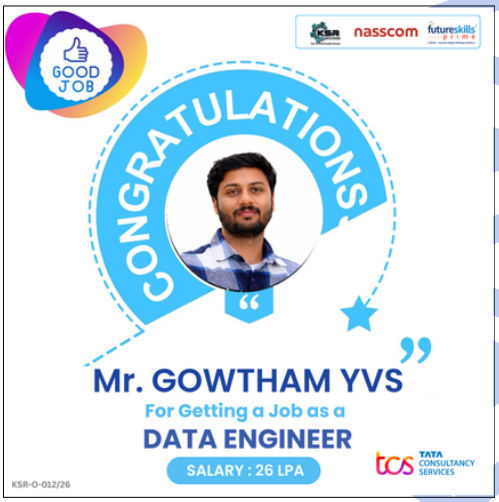




ALUMNI PLACEMENTS (ALL KSR PROGRAMS)





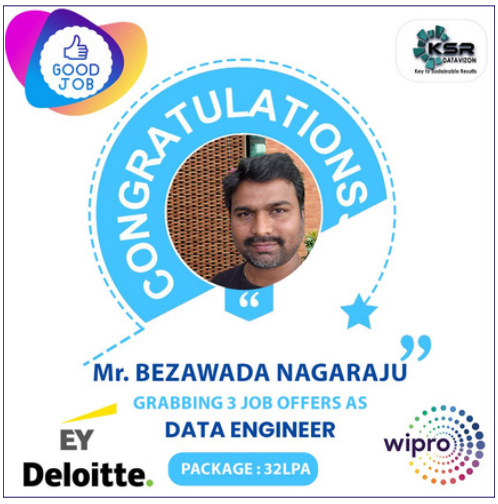














HIRING COMPANIES



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FULL STACK JAVA ROADMAP 2026



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